dGompertz

Parametrisation

The Gompertz distribution with cure fraction has log "survial function"

$$\log S(y) = -\frac{\mu}{\alpha}\left(\exp(\alpha y) - 1\right)$$

for response $y \geq 0$, $\mu > 0$. The "cummulative distribution function" and the "density" then follows as

$$F(y) = 1 - \exp\left[-\frac{\mu}{\alpha}\left(\exp(\alpha y) - 1\right)\right]$$

and

$$f(y) = \mu \exp\left[\alpha y - \frac{\mu}{\alpha}\left(\exp(\alpha y) - 1\right)\right].$$

Note that α is allowed to be negative making to allow for a cure fraction, hence $\lim_{y\to\infty} F(y) < 1$ for $\alpha < 0$.

Link-function

The parameter μ is linked to the linear predictor η as:

$$\mu = \exp(\eta)$$

Hyperparameters

The shape parameter α is represented as

$$\alpha = \theta$$

and the prior is defined on θ.

Specification

- `family="dgompertzsurv"` for survival models
- Required arguments: y (to be given in a format by using `inla.surv()` for survival models)

Hyperparameter spesification and default values

doc `destructive gompertz (survival) distribution`

hyper

theta

hyperid `108101`
name `shape`
short.name `alpha`
output.name.intern `alpha_intern for dGompertz`
output.name `alpha parameter for dGompertz`
initial `-1`
fixed `FALSE`
prior `normal`
param `0 10`
to.theta `function(x) x`

from.theta `function(x) x`

experimental `TRUE`

survival `TRUE`

discrete `FALSE`

link `default log neglog`

pdf `dgompertz`

Example

In the following example we estimate the parameters in a simulated case

```
gen.cure.gompertz = function(n,a,b,p) {
    if (length(b) == 1) b <- rep(b, n)
    if (length(p) == 1) p <- rep(p, n)
    rm = rbinom(n=n,size=1,prob=1-p)
    t=rep(NA,n)
    for(i in 1:n){
        t[i]=ifelse(rm[i]==0,
                    Inf,
                    log((-(a/b[i])*log(1-runif(n=1,min=0,max=1-p[i]))) + 1)*(1/a))
    }
    t_finite = ifelse(t==Inf,0,t)
    u2 = runif(n=n,0,max(t_finite))
    t2 = pmin(t,u2)
    delta = ifelse(t<u2,1,0)
    return(cbind(t2,delta))
}

n = 1000
a_par=-0.5 ##defective
x <- rnorm(n)
b_par <- exp(0.2 + x)
p_par = exp(b_par/a_par) ##cure fraction

data = gen.cure.gompertz(n=n,a=a_par,b=b_par,p=p_par)
colnames(data) = c("tempo", "delta")
data <- as.data.frame(data)

data$x <- x
r <- inla(inla.surv(tempo, delta) ~ 1 + x,
          data = data,
          family = "dgompertzsurv",
          control.inla = list(cmin = 0),
          verbose = TRUE)
summary(r)
```

Notes